



Field study on adaptive thermal comfort in office buildings in Malaysia, Indonesia, Singapore, and Japan during hot and humid season

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ABSTRACT

Thermal comfort studies are necessary in the early stage of building design, especially in tropical climates where buildings are exposed to solar radiation throughout the year. In this study, we aimed to determine the summer comfort temperatures in offices in Malaysia, Indonesia, Singapore, and Japan. Through field surveys during 2015, we collected 2049 responses from 325 occupants in 13 office buildings employing various ventilation modes, namely, free running (FR), mixed mode (MM), and mechanical cooling (CL). The results showed that the comfort range differed for each group of occupants under the different ventilation modes. The comfort operative temperatures in tropical climates are 25.7 °C, 24.9 °C, and 27.5 °C for the CL, FR, and MM ventilation modes respectively, whereas in Japan it is 25.8 °C for both CL and MM. The survey revealed that 97% of the responses in Japan were within the central three points of the TSV scale, but lower percentages were recorded in the other countries. In Singapore, 92.9% of the respondents were feeling either cold, cool, or slightly cool in their offices. In terms of adaptive behaviour, most of the occupants in Malaysia frequently used air-conditioning systems to maintain thermal comfort, while drinking cold beverages was more common in Indonesia and Japan. The high comfort temperature with MM ventilation could be influenced by the strong air movement and a wide range of adaptive options, such as opening windows and utilising air-conditioning systems.

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1. Introduction

It is well known that heating, ventilation, and air-conditioning (HVAC) systems are the main consumers of energy in buildings. Reports from Saudi Arabia [1] and Indonesia [2] indicated that in these countries, air-conditioning (AC) systems consume more than 50% of the total electricity used in buildings. It is highly likely that such consumption would be the same elsewhere. In developing countries with a hot-humid climate, e.g. Malaysia and Indonesia, the use of AC units has increased dramatically over the last ten years. This applies especially to typical modern offices in 'glass box' buildings located in hot-humid climates, where the air-conditioning system is often the only means of ventilation. In

Malaysia [3], the predicted number of AC units in private use is expected to rise to 1.5 million by 2020. Although AC use in Indonesia is currently lower, a survey by Ekasiwi et al. [4] has shown that approximately three-quarters of the respondents started to use AC units in their households between 2005 and 2009. A recent study in Indonesia by Mori et al. [5] has found that people tended to buy AC units as their income increased and they could afford the higher electricity costs.

Obviously, the energy consumption of AC systems increases when lower temperature settings are used in the cooling mode [6], which, in some countries, has led to campaigns encouraging a 'minimum thermostat temperature setting'. The Cool Biz campaign in Japan promotes a minimum AC setting of 28 °C for cooling [7], while, in Malaysia, the government is promoting a minimum thermostat temperature of 24 °C for cooling in governmental offices [8]. Directives on the regulation of thermostat settings are specifically necessary in developing countries situated in a tropical climate. However, various aspects relevant to such regulation have

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to be considered, and it would be prudent to note that the comfort preferences of the population in a certain climatic zone could differ from those of the inhabitants of another climatic zone. Previous field studies relating to various building located in hot-humid climates in Pakistan [9], Cuba [10], Nepal [11], and India [12] have found that in these countries, temperatures of approximately 30 °C were still considered within comfortable temperature ranges, while, in many other places, such temperatures would be considered decidedly uncomfortable.

One of the climatic characteristics in countries considered located in a tropical rainforest climatic zone, such as Malaysia, Indonesia, and Singapore, is the relatively high humidity of between 77% and 88%. Moreover, the annual average air temperature is high at between 20 and 34 °C [13]. In Tokyo, Japan, the average relative humidity could be as high as 78% during the summer season, and the average air temperature in this city is 22–29 °C [14]. However, with the conventional HVAC design, the relative comfortable humidity levels are between 40 and 60% and the air temperature range is between 25 and 27 °C [15].

As the latest HVAC systems are equipped with a dehumidifying feature, the problem of the humidity level has been eliminated in air-conditioned buildings. However, the humidity level remains a significant challenge in the free running (FR) mode when the building is not consuming energy for heating or cooling purposes. A climatic chamber study in Japan by Tsutsumi et al. [16] has indicated that people were more comfortable and productive in conditions of low humidity because of the rapid evaporation of perspiration. On the other hand, people felt tired when the relative humidity reached 70%.

According to the adaptive thermal comfort theory, heat exchange calculations could not be applied strictly to all climatic conditions, as humans tended to adapt to their surroundings [17]. This implies that people overcome their thermal discomfort by adaptive behaviour, such as consuming hot or cold beverages, changing clothes, and opening windows or doors. The adaptive behaviour theory likely explains how people remained comfortable in relatively high indoor temperature conditions, such as the 'set-suden' (saving electricity) conditions of the summer season in Japan [18].

In view of the above, this study aims to investigate the comfort temperature range of office occupants in three tropical climate countries, namely, Malaysia, Indonesia, and Singapore, as well as that during the summer season in Japan, a country with a temperate climate. The results were compared with relevant international standards, namely, the *Comité Européen de Normalisation* (CEN) [19] standard EN15251 for natural ventilation, and the guide on mechanical cooling of the Chartered Institution of Building Services Engineers (CIBSE) (United Kingdom) [20], as well as the local guides from each country. Additionally, as a reference for comfort temperature analysis during this study, the indoor humidity and the air movement were measured and the adaptive behaviours of the occupants during the field survey were observed.

2. Outline of methodology

2.1. Climate description

The annual variation in the outdoor air temperature for each country indicates that the outdoor air temperatures remained relatively constant throughout the year in Malaysia, Singapore, and Indonesia (Fig. 1). The monthly mean temperature in Japan varies significantly along with the seasonal cycle; therefore, the field surveys in Japan were conducted specifically during the summer season. Despite the climatic differences, the monthly mean outdoor air temperature in Japan at the time of the field study was quite

similar to that of Indonesia, with a deviation of only 1.4 °C [14].

2.2. Measurement period

The field survey was conducted during 2015 in the four countries. Some 2049 dataset samples were collected from 325 respondents (Table 1). Each respondent voted 6 to 10 times throughout the study, while the climatic variables were measured during 3–10 working days at each location. The questionnaires were distributed twice a day on the days the measurements were conducted, namely, every morning (between 10:00 and 11:00) and afternoon (between 14:00 and 15:00).

2.3. Building information

Thirteen buildings were investigated in this study (Table 1), namely, four university offices in Malaysia, in UTM Kuala Lumpur and UiTM Shah Alam; three buildings in Indonesia, ID1 in the central area of Bandung, as well as ID2 and ID3 at the ITB campus; two commercial buildings with CL mode in Singapore; and four buildings at TCU in Yokohama and the Setagaya campus. The façades of the buildings are shown in Fig. 2a–g.

In ID3, the ventilation mode is indicated as concurrent MM [21], as the occupants opened the windows while simultaneously using the air-conditioning system during the field measurements. In CL mode, the action of opening windows is considered adaptive behaviour, as it occurred only occasionally. In Japan, both the FR and CL ventilation modes could be categorised as changeover mixed mode, i.e. changing the mode between CL and FR according to the weather [21]. However, to avoid confusion, we separated this specific mode into two different sub-modes according to the usage of mechanical cooling or air-conditioning, i.e. either the CL or the FR mode was employed at the time of measurement.

2.4. Field measurements of climatic variables

Since thermal comfort is affected by physical and personal parameters, both these factors were measured simultaneously in this study. There was no interference in or control over the thermal environment, as this study aims to investigate the response of occupants to typical climatic conditions.

In Malaysia, Indonesia, and Singapore, the air temperature (T_a), globe temperature (T_g), and relative humidity (RH) were measured every 10 s for at least 30 min. During each session, the measurements were conducted while the occupants were answering the questionnaire. The measuring instrument has two external data channels. A 40 mm black sphere was used to measure the globe temperature (Fig. 3a). Simultaneously, the air movement was measured in 10 s intervals by using a hot-wire anemometer.

In Japan, the measurements for each parameter were recorded once for each small group of respondents, 10 min after having set up the instruments. Subsequent to such measurement, the instruments were moved to the next group of respondents.

Each set of instrument and sensor was attached to a retort stand (Fig. 3b) at a height of 1.1 m from the floor level. They were placed approximately 1 m from the respondents. The details for each instrument are provided in Table 2.

The mean radiant temperature (T_{mrt}) and operative temperature (T_{op}) in all the investigated locations were estimated by using equations provided by the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) [15].

2.5. Thermal comfort survey of personal variables

Personal parameters, such as the insulation value of the

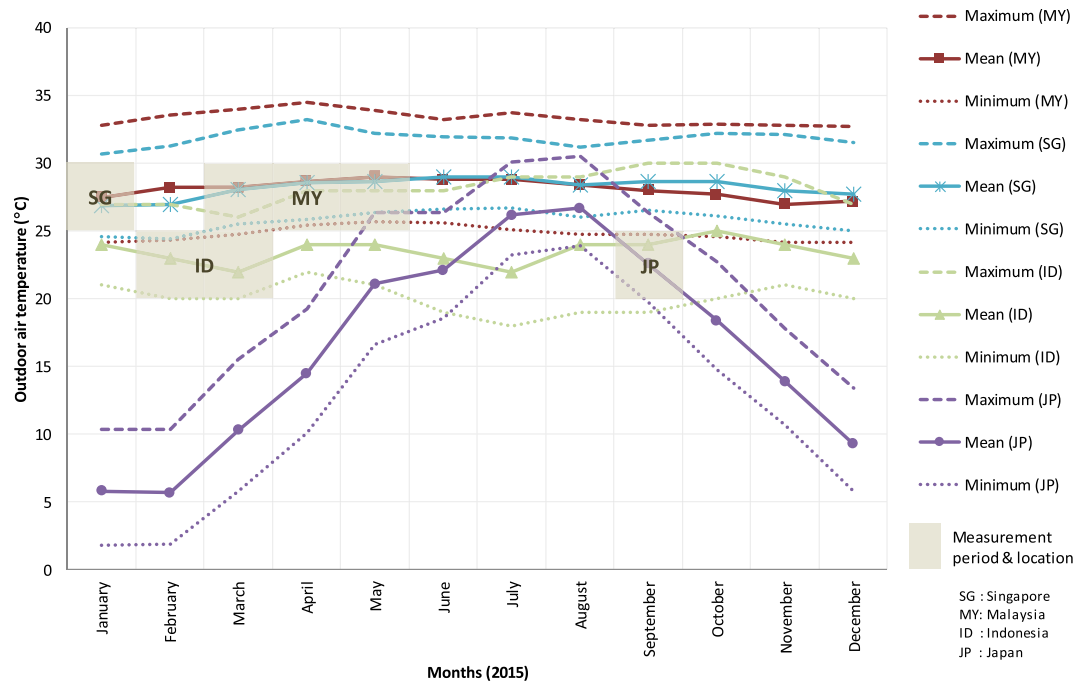


Fig. 1. Annual variation in monthly statistics of outdoor temperature for the year 2015 for each country and measurement timeline. 'MY' refers to the data obtained from the weather data station on the MJIT building at the UTM Kuala Lumpur Campus, Malaysia. 'ID' refers to the data observed from the weather data station in Bandung Airport, Indonesia, approximately 4.5 km to the target location of measurement. 'SG' refers to the data observed from the weather data station in Changi Airport, Singapore, approximately 3 km to the target location of measurement. 'JP' refers to the data observed from the Japan Meteorological Agency Tokyo point, within the Automated Meteorological Data Acquisition System (AMeDAS) network [14], approximately 16 km to the target location of measurement.

Table 1
Information on investigated buildings.

Country	City	Location	Measurement period	Building code	Mode	FL/TF	Orientation	N people	N data
Malaysia	Kuala Lumpur	3°08'N, 101°42'E	13/4/2015–5/5/2015 (20 days)	MY1	CL	4/10	E	15	652
						8/10	N-E	15	
						8/10	S-E	39	
						10/10	E	21	
	Shah Alam (Selangor)	3°04'N, 101°30'E	5/3/2015–21/5/2015 (29 days)	MY2 MY3 MY4	CL	1/2	W	40	486
						1/1	S-W		
						1/1	S-W		
Indonesia	Bandung	6°53'S, 107°36'E	24/2/2015–12/3/2015 (13 days)	ID1 ID2 ID3	FR	2/5	S-N	18	159
					MM	3/3	S-N	20	150
					CL	8/13	N-W	16	91
Singapore	Singapore	1°18'N, 103°50'E	8/1/2015–29/1/2015 (8 days)	SG1 SG2	CL	13/20	S-E	14	56
						4/4	N-E		
Japan	Yokohama	35°33'N, 139°34'E	1/9/2015–25/9/2015 (4 days)	JP1 JP2 JP3 JP4	FR and CL	1/2	S	40	173
						2/2	N		
						5/5	S-N		
						1/4	W-E	87	
	Setagaya (Tokyo)	35°35'N 139°39'E	14/9/2015–18/9/2015 (3 days)		FR and CL	4/4	W		282
Total								325	2049

Note: CL: Cooling, MM: Mixed mode, FR: Free running; FL/TF: Investigated floor level/Total floors of the building; Orientation: S: South; E: East; N: North; W: West; N people: Number of respondents; N data: Number of votes.

occupants' clothing and their thermal perceptions, were surveyed by using the questionnaires. An English questionnaire was distributed in Kuala Lumpur (Malaysia) and Singapore, while, in the other locations, such as Shah Alam (Malaysia), Indonesia, and Japan, the questionnaires were translated into the local languages.

For the thermal sensation vote (TSV) in this study, the ASHRAE seven-point scale was used [22], as shown in Table 3. In Japan, the seven-point scale of the Society of Heating, Air-conditioning, and Sanitary Engineering of Japan (SHASE) was used in the questionnaire. It was used instead of the ASHRAE scale because 'cool' and 'warm' have positive meanings in Japanese translations. These two

sensations could be misinterpreted as comfortable conditions by Japanese people, while they were intended to indicate discomfort in the ASHRAE scale [23]. In addition to TSV, the occupants indicated their thermal preference (TP), which was expressed in the Nicol five-point thermal preference scale [24]. To assess the indoor humidity in the offices, the occupants were asked about their perception (feeling) on humidity (HF) and humidity preference (HP), as indicated in Table 4. In addition, the air movement vote (AV), relevant to a four-point scale, was investigated through the questionnaires. An English version of the questionnaire with Malay translation of clothing items can be found in Appendix A1.



Fig. 2. Investigated buildings in: a) Malaysia; b) Indonesia (MM); c) Indonesia (FR); d) Indonesia (CL); e) Singapore; f) Yokohama, Japan; g) Setagaya, Japan.

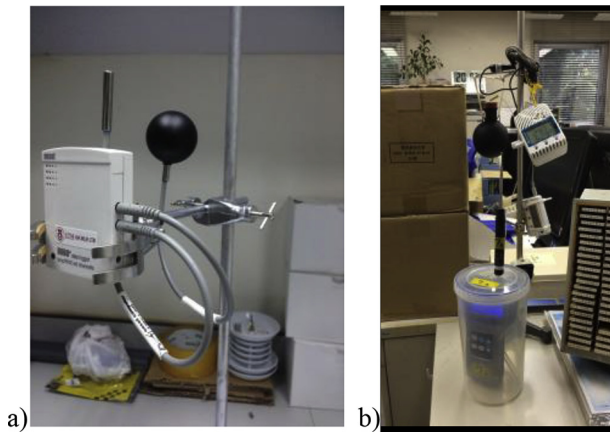


Fig. 3. a) Thermo recorder U12-013, with two external sensors for air temperature, humidity, and globe temperature, used in Malaysia, Indonesia, and Singapore; b) Hot-wire anemometer with probe sensor and thermo recorder on a retort stand, used in Japan.

3. Results and discussions

3.1. Thermal environment during voting

The results from the field measurements and the survey were compiled and the mean values for each parameter were obtained, as indicated in Table 5. The average indoor air temperature displayed here was measured during the field measurement period only, and the result shows that the highest T_a was 26.7 °C in the FR

mode in Indonesia. This value was quite similar to that in the FR mode measured in Japan. In contrast, the lowest T_a was the 23.1 °C measured in Singapore.

As we were using several thermal indices, we noticed that it was important to determine the relation of one thermal index to another. We found that T_a was correlated strongly with the other temperature indices (Table 6 and Fig. 4), which could imply that, to a significant degree, all the thermal indices were in effect measuring the same factor. This confirmation was necessary since T_g was affected by heat radiation, whereas T_{mrt} and T_{op} were estimated based on T_g . These three thermal indices could have different trends if the investigated buildings were not well insulated in extreme climate conditions, which could cause remarkably high heat radiation from the windows, walls, and ceiling. However, since the trend lines in Fig. 4 are almost diagonal, it could be acceptable to use any of these indices in the analysis. In view of this finding, we mainly used T_{op} as the thermal index for further discussion in this study, while the other indices were used only for reference purposes.

The highest relative humidity (RH) was 65% in Singapore in the CL mode, whereas the lowest was 47% in Indonesia in the same mode. The absolute humidity (AH) range was both the lowest at 9.7 g/kg_{DA} in the CL mode in Indonesia and the highest at 13.7 g/kg_{DA} in the FR mode also in Indonesia. The second highest AH at 13.4 g/kg_{DA} was recorded in the FR mode in Japan. Overall, the CL mode showed lower AH compared with that of the FR mode, which was probably the effect of the dehumidifiers of the AC systems. The recent version of the international ASHRAE Standard-55 [25] indicates only an upper limit of 12 g/kg_{DA} for absolute humidity, but does not specify any lower humidity limits.

The average air velocities (V_a) were quite similar in all the locations, ranging between 0.10 and 0.20 m/s, which is a normal condition for an indoor setting.

Table 2
Specifications of the instruments.

Instrument	Manufacturer, Country	Parameters measured	Type of sensor	Resolution	Accuracy and tolerance
Thermo recorder U12-013	Onset HOBO, USA	Air temperature Globe temperature	External sensor cable tmc1-hd External sensor cable tmc1-hd + 40 mm black sphere	0.03 °C	±0.35 °C [0 to 50 °C]
Thermo recorder TR-76Ui	T&D, Japan	Relative humidity Air temperature	Internal sensor External sensor	0.03% RH 0.1 °C	±2.5% RH [10% to 90%] ±0.5 °C
Thermo recorder TR-52i	T&D, Japan	Relative humidity Globe temperature	External sensor	1% RH 0.1 °C	±5% RH [at 25 °C, 50%] ±0.3 °C [−20 to 80 °C]
Hot-wire anemometer	Kanomax, Japan	Air movement	Needle probe 6542-2G	0.01 m/s	±(2% of reading ± 0.0125) m/s [0.10 to 30.0 m/s]

Table 3
Scale for thermal sensation vote and thermal preference.

Scale	Thermal sensation vote, <i>TSV</i>		Thermal preference, <i>TP</i>
	ASHRAE	SHASE	
–3	Cold	Very cold	Much cooler
–2	Cool	Cold	
–1	Slightly cool	Slightly cold	
0	Neutral (<i>neither cool nor warm</i>)	Neutral (<i>neither cold nor hot</i>)	No change
1	Slightly warm	Slightly hot	A bit warmer
2	Warm	Hot	Much warmer
3	Hot	Very hot	

Note: ASHRAE: American Society of Heating, Refrigerating, and Air-Conditioning Engineers; SHASE: Society of Heating, Air-conditioning, and Sanitary Engineering of Japan.

Table 4
Scale for humidity feeling, humidity preference, and air movement vote.

Scale	Humidity feeling (<i>HF</i>)	Humidity preference (<i>HP</i>)	Air movement vote (<i>AV</i>)
–3	Very dry		
–2	Dry	Prefer much drier	
–1	Slightly dry	Prefer slightly drier	
0	Neither humid nor dry	No change	
1	Slightly humid	Prefer slightly more humid	No movement
2	Humid	Prefer much more humid	Weak
3			Moderate
4			Strong

Table 5
Average values for climatic parameters in all countries and ventilation modes.

Country	Mode	Var.	T_a (°C)	T_g (°C)	T_{mrt} (°C)	T_{op} (°C)	RH (%)	AH (g/kg _{DA})	V_a (m/s)
Malaysia	CL	Mean	24.0	24.4	24.8	24.4	58.2	10.9	0.23
	(n = 1115)	S.D.	1.7	1.7	1.8	1.7	5.9	1.3	0.09
Indonesia	FR	Mean	26.7	26.7	26.7	26.7	62.2	13.7	0.07
	(n = 159)	S.D.	0.2	0.2	0.2	0.2	2.3	0.5	0.01
	MM	Mean	26.5	27.1	27.8	27.2	54.3	11.8	0.17
	(n = 150)	S.D.	1.3	1.3	1.2	1.3	4.3	0.9	0.03
	CL	Mean	25.5	25.8	26.2	25.9	47.3	9.7	0.13
	(n = 91)	S.D.	0.4	0.4	0.5	0.4	2.7	0.5	0.03
Singapore	CL	Mean	23.1	23.2	23.3	23.2	64.9	11.6	0.06
	(n = 56)	S.D.	1.3	1.3	1.4	1.3	4.2	1.6	0.03
Japan	FR	Mean	26.6	26.5	26.5	26.5	61.6	13.4	0.22
	(n = 37)	S.D.	0.3	0.3	0.3	0.8	0.4	1.6	0.14
	CL	Mean	26.1	25.9	25.7	25.9	56.7	12.0	0.15
	(n = 418)	S.D.	0.8	0.7	0.9	0.7	6.1	1.5	0.07

Note: CL: Cooling; FR: Free running; MM: Mixed mode; T_a : Indoor air temperature; T_g : Globe temperature; T_{mrt} : Mean radiant temperature; T_{op} : Operative temperature; RH : Indoor relative humidity; V_a : Air movement; AH : Indoor absolute humidity; n: Number of samples; S.D.: Standard deviation.

Table 6
Regression equations and correlation coefficients of air temperature and other thermal indices.

Mode	Items	$T_a:T_g$	$T_a:T_{mrt}$	$T_a:T_{op}$
FR	Eq.	$T_g = 1.03 T_a - 0.89$	$T_{mrt} = 1.08 T_a - 2.22$	$T_{op} = 1.04 T_a - 1.11$
(n = 219)	r	0.99	0.90	0.97
MM	Eq.	$T_g = 0.97 T_a + 1.44$	$T_{mrt} = 0.91 T_a + 3.63$	$T_{op} = 0.96 T_a + 1.82$
(n = 150)	r	1.00	0.98	1.00
CL	Eq.	$T_g = 0.92 T_a + 2.16$	$T_{mrt} = 0.8 T_a + 5.36$	$T_{op} = 0.9 T_a + 2.67$
(n = 1680)	r	0.97	0.84	0.96

Note: CL: Cooling; FR: Free running; MM: Mixed mode; T_a : Indoor air temperature (°C); T_g : Globe temperature (°C); T_{mrt} : Mean radiant temperature (°C); T_{op} : Operative temperature (°C); r: Correlation coefficient; N: Number of samples; Note: all correlation coefficients are significant ($p < 0.001$).

3.2. Thermal sensation and preferences

3.2.1. Distribution of votes

The mean *TSV* values for all the countries and ventilation modes were within the comfort range ($-1 \leq TSV \leq 1$), except for that in Singapore, which, at -1.7 , indicated that the occupants were feeling cold (Table 7). Considering that Singapore had the lowest indoor temperature of all the locations, the mean *TSV* value was possibly

outside the comfortable range. This argument is supported by the mean *TP* value indicated for the same location, which was the only one indicating a preference for a warmer vote (0.2). In all other locations, the average value of the thermal preference of the occupants showed a preference for a cooler condition. The lowest mean *TP* value was -0.3 , which was recorded in the FR mode in both Japan and Indonesia.

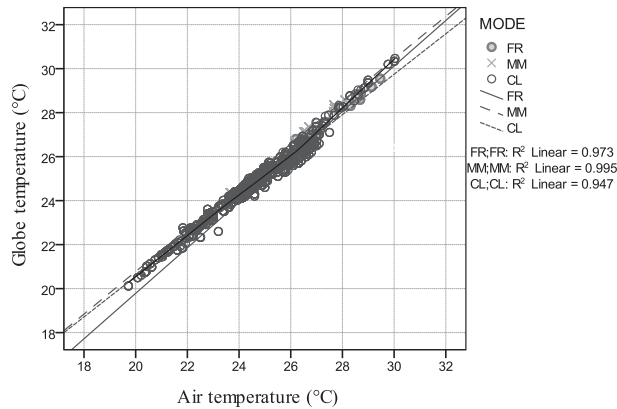


Fig. 4. Correlation of globe temperature and air temperature.

3.2.2. Relation between the thermal sensation and thermal preference

While analysing the relationship between TSV and TP, we found that even when people voted for a 'neutral' sensation ($n = 778$), 19% of them preferred a cooler temperature, whereas 5% preferred a warmer temperature. Fig. 5 shows the cumulative percentages of 'prefer cooler' and 'prefer warmer' votes in relation to TSV. The 'prefer cooler' vote indicates the percentage of occupants who voted for -2 or -1 (much cooler or slightly cooler) on the five-point TP scale, while the 'prefer warmer' vote indicates those occupants who voted for 1 or 2 (slightly warmer or much warmer), as shown in Table 3. Regarding the responses for three warm sensations on the right side of the TSV scale, the proportion of 'prefer cooler' increased while the 'prefer warmer' decreased, and vice versa for the cool sensations. It is apparent that some occupants still preferred an even cooler condition while voting for a cool TSV sensation.

de Dear [26] suggested that at times the thermal preference scale could be a superior measure of the thermal comfort of people, as the 'neutral' condition of TSV was not always considered the preferred condition. In view of a similar study by Rijal et al. [11] relevant to Nepalese people, the inconsistency between the two scales occurred for people who lived in a continuously extremely hot environment, and therefore, preferred a cooler temperature. Alternatively, it could be the natural desire of most people living in a hot climate to prefer a cooler condition, although they probably accepted the prevailing conditions. In all the investigated locations, more than 75% of the votes was within the comfort range of TSV (-1, 0, and 1), except for the FR mode in Indonesia, with 23.9% votes for warm, and the CL mode in Singapore, where, in contrast, 35.7%

Table 7
Average values for thermal sensation vote and thermal preference.

Country	Mode	Var.	TSV	TP
Malaysia	CL ($n = 1115$)	Mean	-0.6	-0.1
		SD	1.0	0.8
Indonesia	FR ($n = 159$)	Mean	1.0	-0.3
		SD	1.4	0.9
	MM ($n = 150$)	Mean	-0.1	-0.1
		SD	1.3	0.8
Singapore	CL ($n = 91$)	Mean	-0.2	-0.1
		SD	1.2	0.7
	CL ($n = 14$)	Mean	-1.7	0.2
		SD	0.9	0.4
Japan	FR ($n = 37$)	Mean	0.4	-0.3
		S.D.	0.6	0.6
	CL ($n = 418$)	Mean	0.1	-0.1
		S.D.	0.6	0.5

Note: CL: Cooling; FR: Free running; MM: Mixed mode; TSV: Thermal sensation vote; TP: Thermal preference; n: Number of samples; S.D.: Standard deviation.

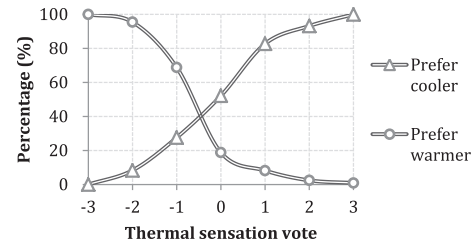


Fig. 5. Relation of TSV and TP scale for all the modes.

of the votes were recorded for cool conditions and 21.4% for cold. In addition to the CL mode in Singapore, the CL mode in Malaysia and the MM in Indonesia recorded the highest percentage vote for -1 or 'slightly cool' (35.9% and 33.3%, respectively). The proportion of occupants who voted for 'neutral' on the seven-point TSV scale was the highest in Japan, at 59.5% and 66.5% for the FR mode and the CL mode, respectively (Table 8).

3.2.3. Thermal comfort zone

The TSV results were analysed by using the probit regression method [27] to predict the thermal comfort zone for the population. Ordinal regression was conducted separately for each country and ventilation mode, with probit as the link function and the operative temperature as the covariate (Table 9). The mean temperatures were calculated by dividing the constant by the regression coefficient, after which each P was calculated by using function (1) and the proportions were plotted, as shown in Fig. 6.

$$\text{Probability} = \text{CDF.NORMAL}(\text{quant}, \text{mean}, \text{S.D.}) \quad (1)$$

where CDF.NORMAL is the cumulative distribution function for normal distribution, 'quant' is the operative temperature ($^{\circ}\text{C}$), with the mean and the S.D. presented in Table 9.

The proportional area of each of the seven-point scale comfort votes were divided by the curves, as shown in Fig. 6a–c. For example, the highest line with square markers in Fig. 6a defines the proportional area for TSV -3 (cold) and TSV -2 (cool), and so forth, until the lowest line of inverted triangles, which defines TSV 2 (warm) above it and TSV 3 (hot) below it. As the operative temperature increased, the proportion of people who voted for a cool sensation diminished and the proportion who voted for a warm sensation increased. We noticed that in Japan (Fig. 6c) there was a probability for only five proportions of thermal votes, as the TSV results were recorded only between -2 and 2.

Using TSV -1, 0, and 1 as the comfortable range, the probits were transformed into proportions, the bell-curve, as shown in Fig. 6d. In all the locations, the optimum proportion of comfortable was relatively high, somewhat over 80% in Malaysia and Indonesia, and

Table 8
Percentage (%) of each thermal sensation vote in each investigated location.

TSV	Malaysia	Indonesia			Singapore	Japan	
	CL	FR	MM	CL	CL	FR	CL
-3	1.4		0.7	1.1	21.4		
-2	17.8	1.9	10.7	12.1	35.7		1
-1	35.9	13.2	33.3	25.3	35.7	2.7	13.4
0	31.9	26.4	30	38.5	7.1	59.5	66.5
1	10.1	18.9	13.3	16.5		35.1	17.7
2	2.5	23.9	6.7	4.4		2.7	1.4
3	0.4	15.7	5.3	2.2			
N	1114	159	150	91	14	37	418

Note: TSV: Thermal sensation vote; CL: Cooling; FR: Free running; MM: Mixed mode.

Table 9
Proportion of TSV over the operative temperature.

Country	Mode	Equation*	Mean (°C)	S.D.	N	R ²	S.E.
Malaysia	CL	$P(\leq -3) = 0.211 T_{op} + 2.86$ $P(\leq -2) = 0.211 T_{op} + 4.24$ $P(\leq -1) = 0.211 T_{op} + 5.28$ $P(\leq 0) = 0.211 T_{op} + 6.35$ $P(\leq 1) = 0.211 T_{op} + 7.20$ $P(\leq 2) = 0.211 T_{op} + 8.01$	13.6 20.1 25.0 30.1 34.1 37.9	4.736	150	0.104	0.019
Indonesia	MM	$P(\leq -3) = 0.573 T_{op} + 12.38$ $P(\leq -2) = 0.573 T_{op} + 14.06$ $P(\leq -1) = 0.573 T_{op} + 15.43$ $P(\leq 0) = 0.573 T_{op} + 16.42$ $P(\leq 1) = 0.573 T_{op} + 17.03$ $P(\leq 2) = 0.573 T_{op} + 17.51$	21.6 24.6 26.9 28.7 29.7 30.6	1.746	1114	0.315	0.079
Japan	CL	$P(\leq -2) = 0.104 T_{op} + 0.34$ $P(\leq -1) = 0.104 T_{op} + 1.62$ $P(\leq 0) = 0.104 T_{op} + 3.56$ $P(\leq 1) = 0.104 T_{op} + 4.88$	3.2 15.6 34.4 47.0	9.640	418	0.004	0.078

Note: $P(\leq -3)$ is the probit of the proportion of the votes that are -3 and less; $P(\leq -2)$ is the probit of the proportion that are -2 and less, and so forth; S.D.: standard deviation; N: number of samples; R²: Cox-Snell R²; S.E.: standard error of the regression coefficient; * regression coefficient is significant (Indonesia and Malaysia: $p < 0.001$; Japan: $p < 0.05$).

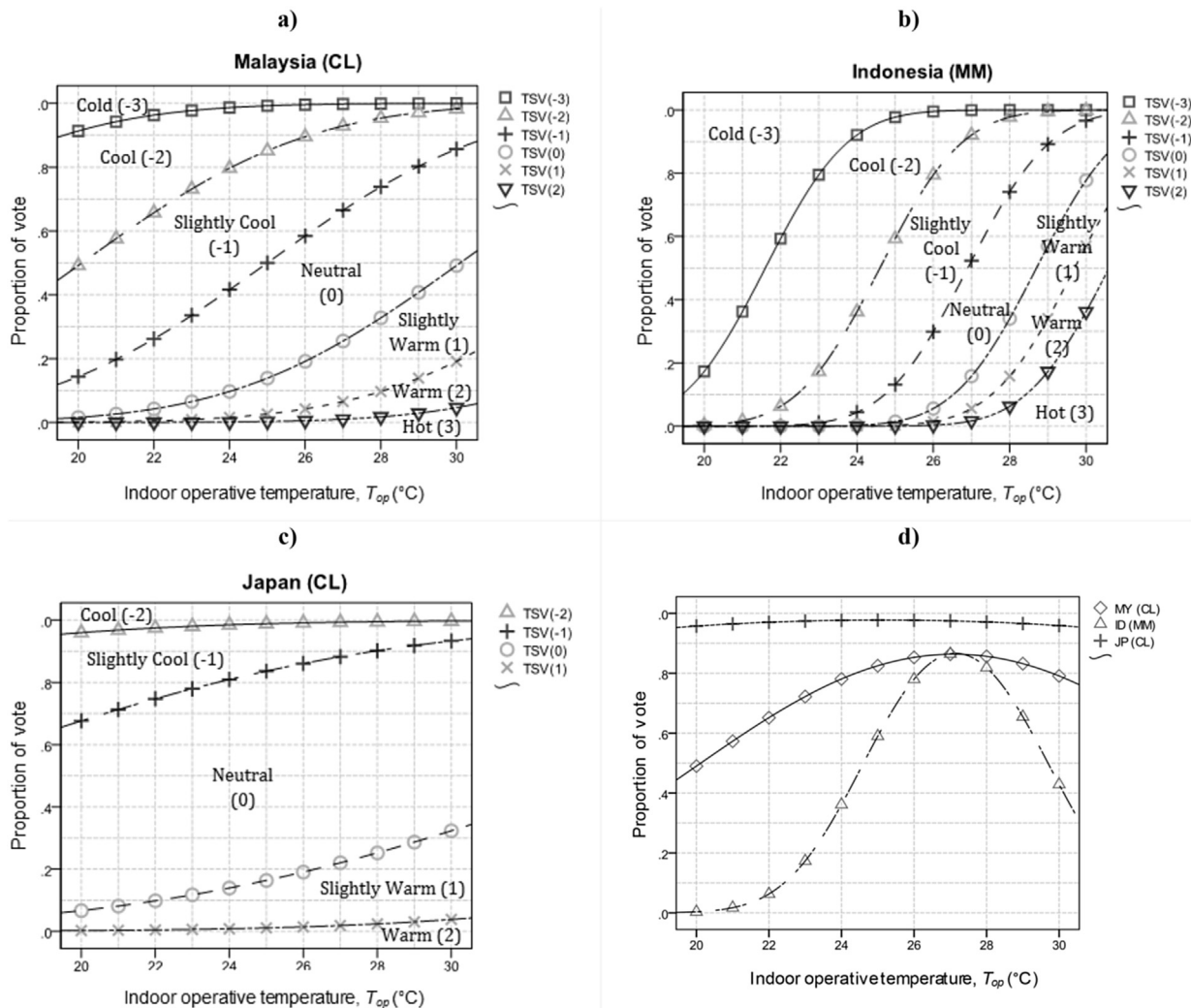


Fig. 6. Proportion of TSV: a) Malaysia CL mode; b) Indonesia MM mode; c) Japan CL mode; d) proportion of comfortable vote.

only slightly less than 100% in Japan. The proportion of comfortable in Japan was quite flat, as 97.6% of people voted either -1 , 0 , or 1 , which meant that nearly all the respondents were thermally

comfortable. More than 80% of people were comfortable at a temperature range of approximately 26 – 28 °C in Indonesia, whereas, in Malaysia, the range was between 24.5 and 30 °C, with the

optimum being 27 °C. This finding indicates that the comfort temperature range in the CL mode in both Malaysia and Japan was wider than with the MM ventilation in Indonesia. However, the comfort temperature with an optimum proportion of people (less than 90%) was similar, at approximately 27 °C. The optimum percentage was lower in comparison with the results of a previous study in residential Japan, where almost 100% of the occupants felt comfortable at temperatures of approximately 24–26 °C. However, this result could be ascribed to people probably adapting better in their own dwellings than in an office environment [23].

3.3. Comfort temperature

3.3.1. Griffiths' method

In this study, the variation in the results of the climatic measurements was quite narrow in some locations. Therefore, the Griffiths' method was deemed suitable to calculate the comfort temperature, based on the following equation [17, 28–30].

$$T_c = T + \frac{(0 - C)}{a} \quad (2)$$

where T_c indicates the comfort temperature (°C), based on T , which is temperature (°C); C is the thermal sensation vote on the scale, where 0 indicates a neutral condition; and a indicates the constant rate of thermal sensation change with the room temperature. In this study, 0.5 was used as the constant, as had been applied by Humphreys et al. [29], with a seven-point thermal sensation scale.

3.3.2. Mean comfort temperature

The comfort temperatures were obtained from this equation for each mode in the four thermal indices, namely, T_a , T_g , T_{mrt} , and T_{op} , as shown in Fig. 7. As indicated by Table 10, the comfort temperatures were compared with the average temperatures when the occupants voted for $TSV = 0$, or 'neutral', $TP = 0$, or 'no change', and the overall comfort ($OC = 5$ or 6, which indicated 'comfortable' and 'very comfortable'. The results of the Griffiths' method relevant to the comfort temperature were similar to all the average temperatures, especially with the CL mode in Japan and MM ventilation in Indonesia.

The mean comfort temperature with MM ventilation in Indonesia at 27.5 °C comfort T_{op} was the highest value, whereas the lowest was also recorded in Indonesia in the FR mode at 24.7 °C. The average comfort temperature in the CL mode in Indonesia was 26.3 °C, identical to the result with the same ventilation mode in Singapore, which was 26.4 °C. The results with the CL mode in Malaysia and Japan were slightly higher, but were comparable at 25.6 °C and 25.8 °C, respectively.

In view of the results in Indonesia, the differences between the comfort temperature levels could be caused by differences in the ventilation mode. However, Fig. 7 and Table 10 show that in both the FR and the CL modes, the mean comfort temperature in Japan was 25.8 °C. The similarity of the two different modes in Japan does not necessarily invalidate the results from Indonesia, as the respondents for each mode in Indonesia were different people, whereas, in Japan, they were the same people. As explained in Section 2.3, changeover mixed-mode ventilation was used in the buildings in Japan; therefore, the occupants changed the mode to suit their needs, usually in relation to the seasonal temperature changes. The comfort temperature result in Japan could probably be a relatively constant temperature setting, regardless of the ventilation mode people were using. This setting might not be suitable for completely different outdoor climatic conditions; however, a recent study in Japan has shown that the comfort temperature of Japanese office occupants was 24.3 °C in the winter

season, which was only 1.1 °C lower than that in the summer season [31].

In hot and humid weather, a result with high comfort temperature (e.g. similar to outdoor temperature) usually meant there would be potential for energy saving, as employing mechanical cooling could be avoided. However, there is concern about wasting energy related to concurrent MM ventilation, as mechanical cooling and operable windows are used at the same time and space [32]. The energy demand could be high because of the outside air entering through the open windows, forcing the AC thermostat to operate longer.

While the occupants in building I3 were not asked about the reason for opening the windows during CL mode operation, the design of office I3 could be a contributing factor. These design aspects in office I3 could include the curtain-wall façade at the south-west, which caused solar radiation to the office. In addition, the AC outlets were located on the 3.6-m-high ceiling, which could have resulted in inadequate cooling. In addition to these design aspects, we noticed that the windows were always open, whereas the AC only started operating at approximately 10:00 when the field survey started. The AC was switched off before the occupants left the office in the afternoon. The mean T_o during the field survey sessions in office I3 was quite high, namely, 28.6 °C during the morning session and 28.9 °C in the afternoon. Meanwhile, the average outdoor air velocity was approximately 14 km/h daily, and the maximum could reach 28 km/h, as was recorded by the weather data station in Husein Sastranegara airport on the measurement days. This windy condition despite the high outdoor temperature could be one of the reasons for occupants opening the windows, since it allowed evaporative cooling.

3.3.3. Comparison with existing research

As shown in Table 11, the results of the current study were compared with the results of similar investigations in several other countries on the thermal comfort relevant to hot climatic conditions. Based on the current study, the comfort operative temperature in Singapore was two degrees higher than was that indicated by de Dear et al. [33]. These authors estimated that the comfort operative temperature in air-conditioned buildings was 24.2 °C. The comfort temperature in Malaysia indicated by the current study was quite similar to that of office spaces in a hospital building in the same country, which was an effective temperature of 25.0 °C [34]. The FR mode in Japan was almost identical to the comfort globe temperature of 25.6 °C in a naturally ventilated office, as indicated by a previous study by Indraganti et al. [18]. However, this temperature was significantly lower than was the temperature relevant to mechanical cooling indicated by the results of the same study. The trend found in the current study is similar to that found by a field study conducted in Fukuoka by Mustapa et al. [35], who found that the comfort temperatures of Japanese people remained constant, regardless of the ventilation modes they employed. Compared with the current study, the comfort temperatures of the residents in Fukuoka were slightly higher, at 26.6 °C in the FR mode and 26.5 °C in the CL mode. The result of the current study is identical to that of a previous thermal comfort study by Karyono [36], who estimated a comfort temperature of 24.7 °C in a university classroom in Bandung. The MM comfort temperature of the current study was slightly higher than was that of a similar study in Malaysia, which estimated a comfort operative temperature of 26 °C in an air-conditioned office, where 50% of the occupants opened the windows [37].

3.4. Comparison with related standards

3.4.1. Free running mode

Various international standards are available for predicting the

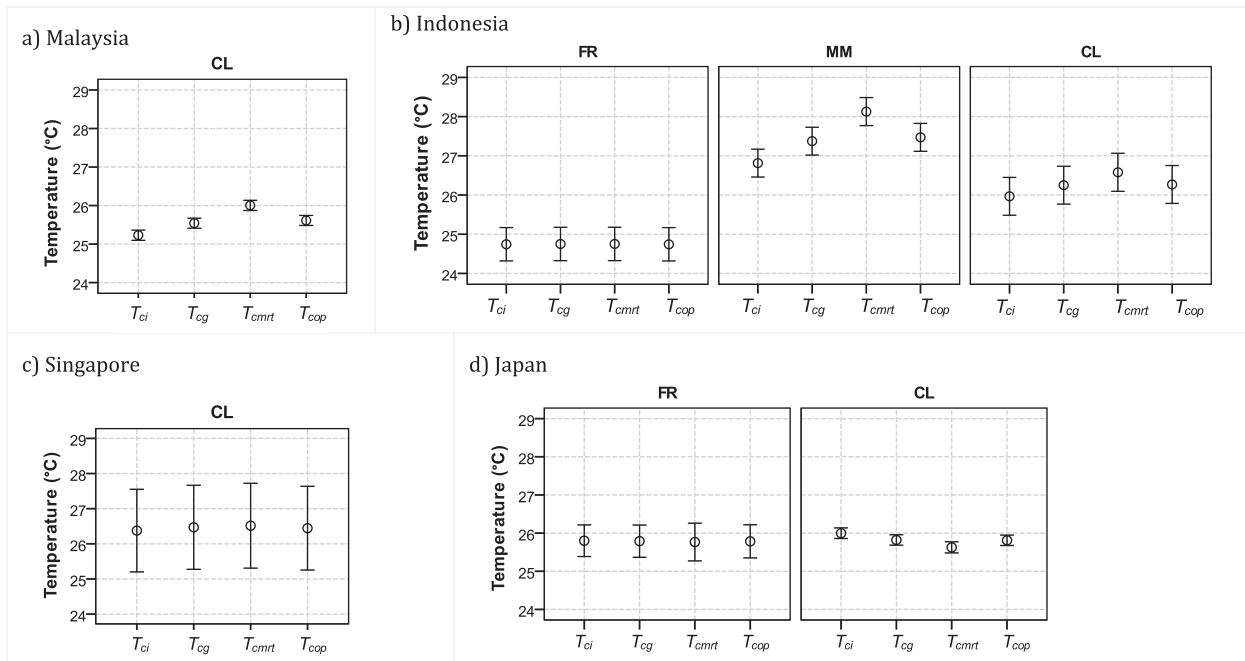


Fig. 7. Mean comfort temperature, with 95% confidence interval, in each location, based on Griffiths' method.

Table 10

Griffiths' comfort temperature and mean operative temperatures for given votes.

Country	Mode	Griffith's method			TSV = 0			TP = 0			OC = 5 or 6		
		N	Mean	S.D.	N	Mean	S.D.	N	Mean	S.D.	N	Mean	S.D.
Malaysia	CL	1114	25.6	2.2	355	24.6	1.7	515	24.2	1.6	823	24.3	1.5
Indonesia	FR	159	24.7	2.7	42	26.7	0.2	65	26.7	0.2	87	26.7	0.2
	MM	150	27.5	2.2	45	27.4	1.2	71	27	1.3	105	27.2	1.2
Singapore	CL	91	26.3	2.3	35	25.8	0.4	60	25.8	0.5	81	25.9	0.5
	CL	14	26.4	2.1	1	24.6	n/a	11	23.3	1.4	11	23.3	1.4
Japan	FR	37	25.8	1.3	22	26.6	0.4	23	26.6	0.5	4	26.5	0.7
	CL	418	25.8	1.4	278	25.9	0.7	320	25.9	0.7	86	25.9	0.8

Note: FR: Free-running, MM: Mixed-mode, CL: Cooling, TSV: Thermal sensation vote, TP: Thermal preference, OC: Overall comfort, N: Number of sample, S.D.: Standard deviation.

indoor comfort temperature, based on the outdoor temperature. To check the fit of the results of the current study with related standards and guidelines, comfort operative temperatures were plotted against the running mean temperatures (Fig. 8). Each point represents the average value of the comfort operative temperatures for each occupant.

As the field measurements for the current study were conducted daily, the results for the FR mode were compared with Equation (3) from CEN EN15251, which predicts the comfort temperature zones in the FR mode, based on the daily running mean temperature [19].

$$T_{comf} = 0.33T_{rm} + 18.8 \quad (3)$$

where T_{comf} is the comfort temperature (°C), and T_{rm} is the daily running mean temperature (°C), which is based on the outdoor temperature data sets of the seven days prior to the day of calculation [19].

In addition to the standards and guidelines, local regulations were taken into account for comparison with results from each investigated location. In Japan and Malaysia, the minimum thermostat settings based on the energy saving campaigns were 28 °C and 24 °C, respectively [7,8]. In Indonesia, the national standard SNI 6390:2011 recommends an indoor air temperature of 25.5 °C [43].

In Singapore, according to Standard SS 554:2009, the comfortable indoor operative temperature is between 24 and 26 °C [44]. The comparison of the obtained comfort temperatures and the local regulations in each country is summarised in Table 12.

The results for the FR mode in both Indonesia and Japan were mostly within the comfort zone of the CEN standard category I ($\pm 2K$), as indicated in Fig. 8a. However, in Indonesia, some points were below the lower limit of category III ($-4K$), whereas only two points in Japan were over the upper limit of category II ($+3K$). These results could be ascribed to the narrow range of T_{rm} , which causes variations in the comfort temperature for similar daily running mean temperatures.

We use EN 15251 standard since it is based on the daily running mean outdoor temperature, whereas the ASHRAE standard (2004) [45] is based on the monthly mean outdoor temperature. Even though the modified ASHRAE standard (2013) [25] indicated the daily running mean outdoor temperature, the equation is based on the monthly mean outdoor temperature. The EN 15251 standard is more appropriate than the ASHRAE standard to indicate the daily variation in comfort temperature. For reference purposes, the individual mean comfort temperatures in the FR mode were also plotted against the monthly mean outdoor temperature, compared

Table 11

Comparison of comfort temperatures of the current study with those of previous studies.

Year of survey	Reference	Country	Mode	Thermal index	T_{comf} (°C)	S.D. of T_{comf} (°C)
1986	de Dear et al. [33]	Singapore	CL	T_{op}	24.2	
1999	Cena et al. [38]	Australia	CL	T_{op}	23.3	
2000	Karyono [39]	Indonesia	CL	T_{op}	26.7	
2011	Azizpour [34]	Malaysia	CL	ET	25.0	
2012	Indraganti et al. [18]	Japan	CL	T_g	27.1	
2014	Mustapa et al. [35]	Japan	CL	T_{op}	26.5	
2015	This study	Singapore	CL	T_{op}	26.4	2.1
2015	This study	Indonesia	CL	T_{op}	26.3	2.3
2015	This study	Malaysia	CL	T_{op}	25.6	2.2
2015	This study	Japan	CL	T_g	25.8	1.4
1988	Busch [40]	Thailand	FR	SET	22.8	
2005	Karyono [36]	Indonesia	FR	T_a	24.7	
2012	Indraganti et al. [18]	Japan	FR	T_g	25.6	
2014	Mustapa et al. [35]	Japan	FR	T_{op}	26.6	
2015	This study	Indonesia	FR	T_a	24.7	2.7
2015	This study	Japan	FR	T_{op}	25.8	1.3
2010	Deuble et al. [41]	Australia	MM*	T_{op}	22.1	
2009	Daghigh et al. [37]	Malaysia	MM**	T_{op}	26.0	
2015	This study	Indonesia	MM**	T_{op}	27.5	2.2

Note: CL: Cooling; FR: Free running; MM*: Change-over mixed mode; MM**: Concurrent mixed mode; T_a : Indoor air temperature (°C); T_g : Globe temperature (°C); T_{op} : Operative temperature (°C); T_{mrt} : Mean radiant temperature (°C); ET : Effective temperature (°C); SET : Standard effective temperature (°C); T_{comf} : Comfort temperature (°C); S.D.: Standard deviation.

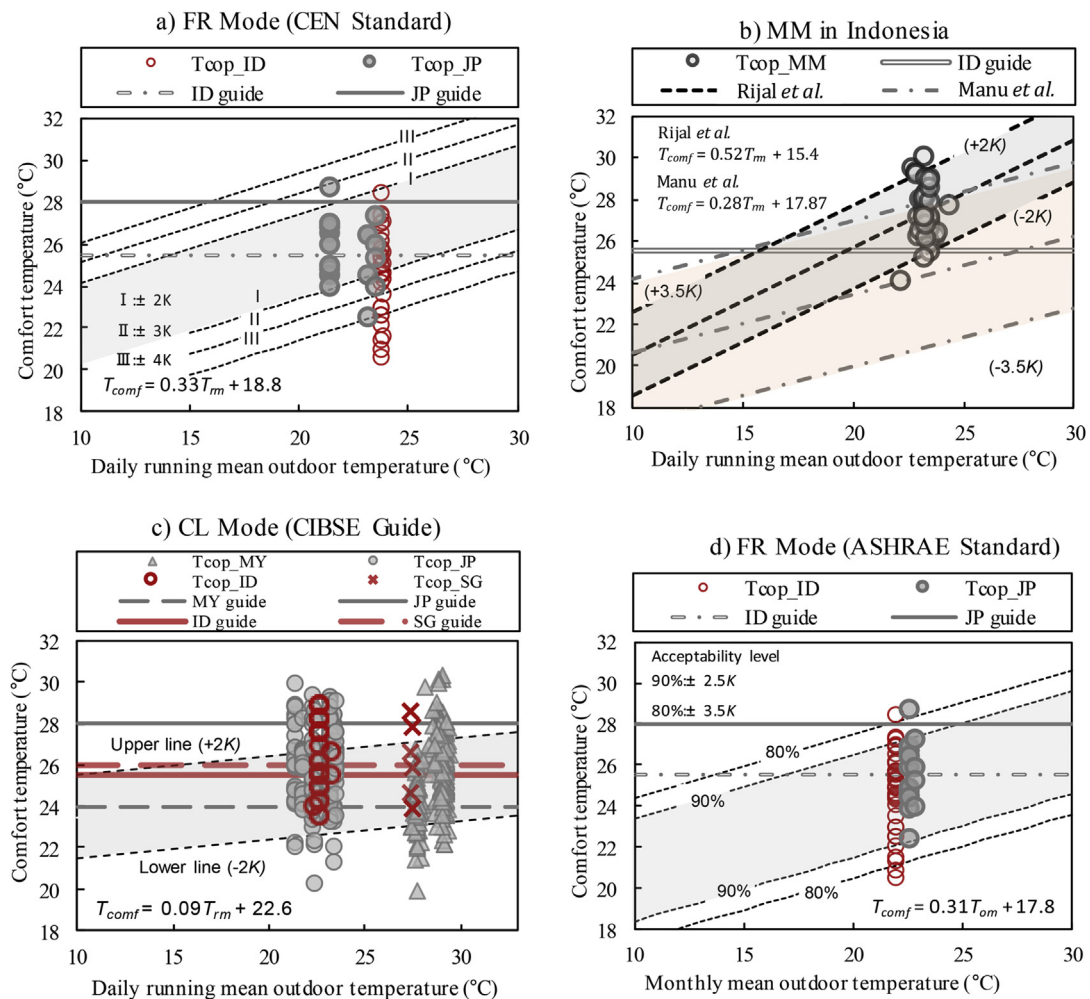


Fig. 8. Comparison of comfort temperature results with standards: a) FR with the CEN standard, b) MM with other studies, c) CL with CIBSE guide, d) FR mode with the ASHRAE standard.

Table 12

Thermal comfort zone in each investigated location based on current study and local regulations.

Country	Mode	T_{cop} (°C)		Local regulation	
		Mean	S.D.	Source	Temperature (°C)
Malaysia	CL	25.6	2.2	Malaysian government [8]	24.0
Indonesia	FR	24.7	2.7	SNI 6390:2011 [43]	25.5
	MM	27.5	2.2		
	CL	26.3	2.3		
Singapore	CL	26.4	2.1	SS 554: 2009 [44]	24.0–26.0
Japan	FR	25.8	1.3	Cool Biz campaign [7]	28.0
	CL	25.8	1.4		

Note: FR: Free-running, MM: Mixed-mode, CL: Cooling, T_{cop} : Griffiths' comfort operative temperature, S.D.: Standard deviation.

with the ASHRAE standard in Fig. 8d.

3.4.2. Mixed mode

Currently, there is no standard relevant to the comfort temperature zone in mixed-mode ventilated buildings. For comparison, the results from the current study were plotted with Equations (4) and (5). The two equations are adaptive models from previous field studies of mixed-mode ventilated buildings during hot-humid weather in Pakistan by Rijal et al. [46] and in India by Manu et al. [12] respectively.

$$T_{comf} = 0.52T_{rm} + 15.4 \quad (4)$$

$$T_{comf} = 0.28T_{rm} + 17.9 \quad (5)$$

As shown in Fig. 8b, the results from the MM samples of the current study fit better with the Pakistani database. The equation line from the Pakistani database intersects the middle part of the results dataset of the current study, whereas the Indian equation only intersects the lowest point of the current study. Both the upper and lower lines of the Pakistani adaptive model were plotted to fit the 90% acceptance category ($\pm 3K$). Compared with the Indonesian guideline, most of the MM results from the current study were above the recommended line, which differs from the FR and CL modes in Indonesia.

3.4.3. Cooling mode

In addition, there is no international adaptive standard for comfort temperature in HVAC buildings, because, generally, the infiltration of outdoor air into such buildings is assumed minimal. However, a correlation was still found between the outdoor and indoor air temperatures in HVAC buildings [47], although this correlation was not as significant as that for the FR modes. Therefore, the comfort zones for the CL mode were plotted with Equation (6) from the CIBSE guidelines, specified for mechanical cooling and heating [20], with $\pm 2K$ upper and lower limits, as shown in Fig. 8c.

$$T_{comf} = 0.09T_{rm} + 22.6 \quad (6)$$

The results of the CL mode were spread within and outside the CIBSE comfort zone. It is noticeable that the results from Indonesia and Singapore were either within or above the comfort zone, with none being below it. However, the results from both these countries had an extremely narrow T_{rm} range, with the local guidelines from each country running through the middle points of the results. This indicates that the local guidelines in Indonesia and Singapore were compatible with the results of the CL mode of the current study. However, as regards the indoor climatic data and the TSV results of the respondents in Singapore, it appears that the recommended temperature setting was not applied at the investigated site.

On the other hand, the results from Japan and Malaysia showed

more variation and were both above and below the comfort range. Most of the results from Japan were below the Japanese guideline, while more than half of all the results from Malaysia were above the Malaysian guideline. This could imply that the Japanese guideline overestimated the comfort temperature, whereas the Malaysian guideline underestimated the comfort temperature of some people in the country, or at least those included in the sample of this study.

3.5. Effect of humidity on comfort temperature

The relatively high average humidity is assumed to have an effect on the thermal comfort of occupants. Overall, the mean HP value corresponded with the HF value (Table 13). However, these subjective votes on humidity were not related with the actual RH and AH . The average HF value pointed to a neutral feeling in the CL mode in Malaysia, while a dry condition was indicated for all the modes in Indonesia. In contrast, the mean HF of the Japanese occupants in all the modes showed that they tended to feel a humid sensation. To account for this, humidity analysis was only performed on the results from Japan.

Table 14 shows how the perception of occupants on humidity correlate with the comfort temperature. The results show that comfort operative temperatures are significantly correlated with HF and HP . This indicates that the comfort temperature (T_{cop}) is high when occupants feel 'less humid'.

Further analysis was conducted by using a simplified three-point HP scale: '-1, prefer drier', '0, prefer no change', and '1, prefer more humid'. Fig. 9 shows the plot of average T_{cop} values for each HP sensation vote in the FR and CL modes in Japan. The lowest T_{cop} values were obtained when the occupants voted '-1, prefer drier'. Occupants generally prefer low humidity conditions that allow more heat loss from evaporation in summer [16].

While humidity (RH , AH) had no significant effect on comfort temperatures, there is a significant relationship between subjective humidity votes and comfort temperatures. This is similar to the findings by Rijal et al. [23], who used 8000 samples from a residential building, which showed that neither absolute nor relative humidity had any significant effect on the comfort temperatures of the occupants. However, the results indicated that when the occupants felt moisture on their skin, their comfort temperatures dropped by approximately 1.3 K. This implies that while the actual climatic parameters had no effect on objective measurements, there was a correlation between the latter and the subjective votes. It is possible that neither RH nor AH could be used to describe the occupants' perception of humidity or skin moisture. This study also demonstrated that in the FR mode, the comfort temperatures were higher by 2 K when the occupants voted 'prefer dry condition' than when they voted 'prefer wet condition'.

Table 13

Average values for humidity feeling and humidity preference.

Country	Mode	Var.	HF	HP
Malaysia	CL	Mean	0.0	0.1
	(n = 1115)	SD	0.9	0.7
Indonesia	FR	Mean	-0.2	0.3
	(n = 159)	SD	0.8	0.6
	MM	Mean	-0.3	0.1
	(n = 150)	SD	0.8	0.6
	CL	Mean	-0.2	0.2
	(n = 91)	SD	0.6	0.5
Japan	FR	Mean	0.4	-0.2
	(n = 37)	S.D.	0.9	0.8
	CL	Mean	0.2	-0.1
	(n = 418)	S.D.	0.9	0.6

Note: CL: Cooling; FR: Free running; MM: Mixed mode; HF : Humidity feeling; HP : Humidity preference; n: Number of samples; S.D.: Standard deviation.

Table 14

Correlation coefficient of humidity subjective votes and comfort temperature of occupants in Japan.

Mode	Items	$T_{cop}:HF$	$T_{cop}:HP$
FR ($N = 37$)	R	−0.326	0.381
	P	0.05	0.02
CL ($N = 418$)	R	−0.263	0.262
	P	<0.001	<0.001

Note: r : Correlation coefficient; p : Significance level; N : Number of samples; T_{cop} : Comfort operative temperature ($^{\circ}\text{C}$); HF : Humidity feeling; HP : Humidity preference.

proportional gap between this action and the rest of the adaptive actions. The results of the current study are similar to those of the study by Rijal et al. [46] conducted in Pakistan, indicating that when mechanical cooling was available, the tendency was to not open the windows, especially when the outdoor temperature was high. On the other hand, some occupants in Malaysia informed us that they sometimes opened the windows to let in warmer air from outside when the indoor temperature was too cold. Obviously, this pointed to an instance of the overcooling of a building in a hot-humid climate.

In both Indonesia and Japan, the most frequent adaptive action

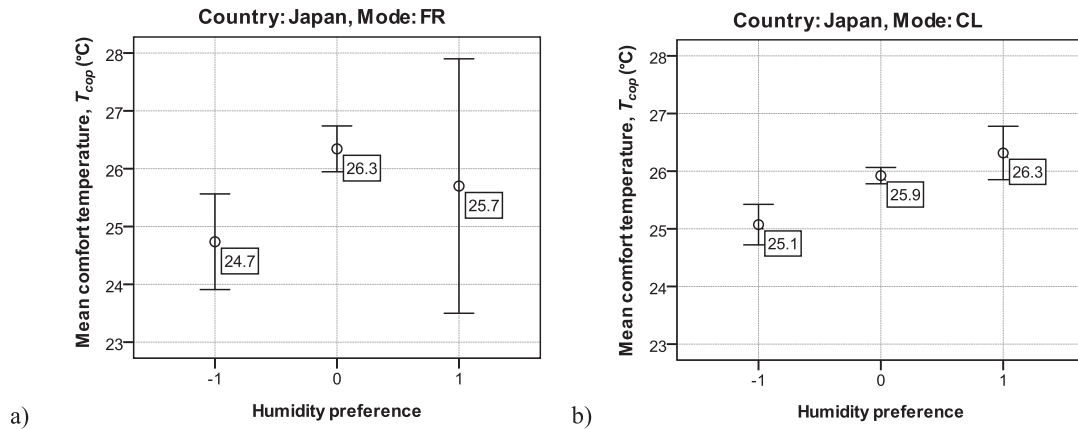


Fig. 9. Mean comfort temperatures relevant to humidity preference of occupants in Japan: a) FR mode, b) CL mode.

3.6. Effect of air movement on comfort temperature

In addition to humidity, indoor air movement affected the comfort temperature. All locations showed AV values greater than 2.0, indicating ‘weak air movement’, except in Indonesia with a lowest mean AV of 1.8, because of the average actual air movement being below 0.10 m/s (Table 15). The highest AV votes of 2.7 were recorded in Singapore and Japan. The actual air movement inside the office spaces and the perception of air movement by the occupants could have affected their thermal sensation vote.

The results show that comfort temperatures increase with air movement. The trends illustrated in Fig. 10 confirm that higher AV will result in higher comfort temperatures, which is important in MM ventilation. Conversely, the low comfort temperatures in the FR mode could be caused by a lack of air movement. Similar results were shown by Cândido et al. [48] in northeast Brazil, where comfort levels were close to the daily average outdoor temperatures because of adequate air movement. Nicol [49] postulated that comfort temperature could be increased by up to 4 $^{\circ}\text{C}$ if the air movement were 1 m/s.

3.7. Adaptive behaviour

The questionnaire survey included a multiple-choice question on which methods the occupants had employed recently to overcome their heat discomfort.

Regardless of the ventilation modes, the adaptive behaviour of the respondents in each country was different. Fig. 11 shows the proportion of each adaptive action in each country. The data were collected in binary mode, where ‘1’ represented a vote for the action, and the options not selected were input as ‘0’. In Malaysia, the most frequent action was switching on the AC, according to the 47.5% vote of the respondents. However, there was a significant

Table 15

Average values for air movement votes.

Country	Mode	Var.	AV
Malaysia	CL ($n = 1115$)	Mean	2.4
		SD	0.7
Indonesia	FR ($n = 159$)	Mean	1.8
		SD	0.8
	MM ($n = 150$)	Mean	2.1
		SD	0.7
Singapore	CL ($n = 91$)	Mean	2.2
		SD	0.5
	CL ($n = 14$)	Mean	2.7
		SD	0.5
Japan	FR ($n = 37$)	Mean	2.7
		S.D.	0.7
	CL ($n = 418$)	Mean	2.6
		S.D.	0.8

Note: CL: Cooling; FR: Free running; MM: Mixed mode; AV: Air movement vote; n : Number of samples; S.D.: Standard deviation.

was to eat or drink something cold, indicated by the high proportions of 64.3% and 78.7%, respectively. The second most frequent adaptive action was opening windows or doors in Indonesia, with 22.6% of the votes. A study by Brager et al. [50] found that people having more control over operable windows had 1.5 $^{\circ}\text{C}$ higher comfort temperatures compared with those who had no control or access to such windows. This is a reasonable explanation for the higher comfort temperatures in the MM mode, where the occupants had control over and access to both the windows and the AC system. In Japan, the second most favoured adaptive action was using a pedestal fan (6.8% of the votes).

In contrast with Japan, pedestal fans or smaller personal fans are not common in Bandung, Indonesia. In all the investigated buildings in Indonesia, we never saw the occupants using a fan as a means of cooling down, which is in stark contrast to the people in Japan. A lesser preferred adaptive action was adjusting clothing,

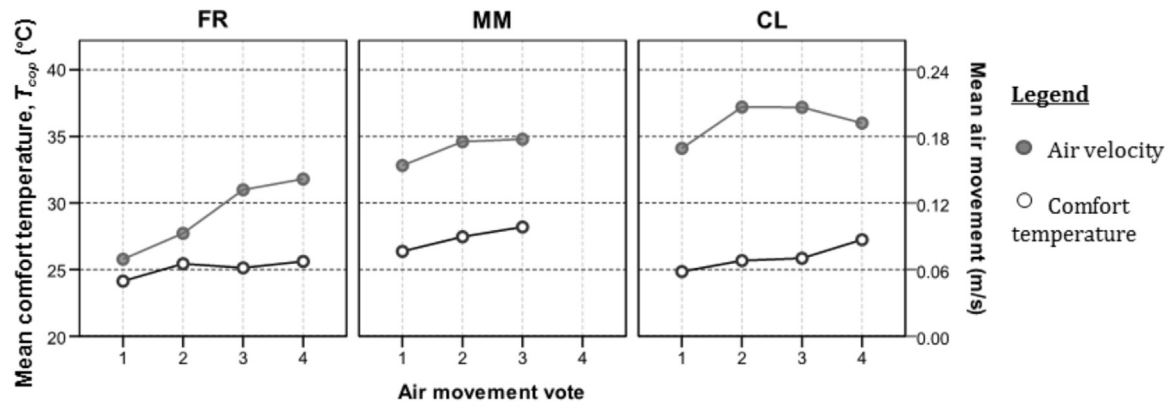


Fig. 10. Mean comfort temperature and mean air movement relevant to air movement vote.

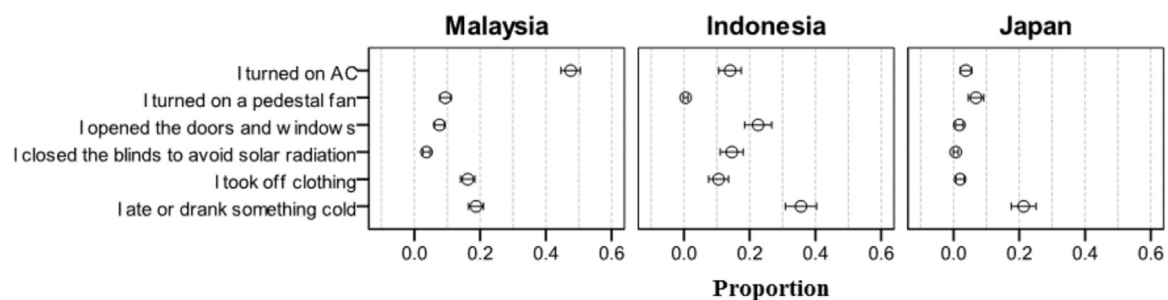


Fig. 11. Proportion of occupants' adaptive behaviour in Malaysia, Indonesia, and Japan.

which is comparable with the findings of Karyono [51]. He found that in Indonesia, adjustment of clothing was more likely applied during cool weather than during warm weather.

In this survey, personal parameters were also collected from the respondents. Their clothing insulation averages (I_{cl}) have similar values across locations, as shown in Table 16. The highest mean I_{cl} was recorded in Malaysia (0.62 clo) and the lowest was recorded in Japan (0.51 clo). Goto *et al.* [52] and Tanabe *et al.* [53] have shown that Japanese people tended to adjust their clothing according to the outdoor temperature of the previous day. This tendency is not seen in occupants in hot-humid climates—consistent weather leads to lack of clothing adjustment and the correspondingly low correlation between outdoor temperature and clothing adjustment. This is evidenced by Malaysia having the highest mean I_{cl} despite having higher outdoor temperatures than Japan (see Fig. 1). This could be related to high air-conditioning usage, as well as cultural and social norms. In Japan, the Cool Biz campaign has tried to eliminate such other factors by encouraging office workers to wear thermally comfortable clothes during summer [42].

4. Conclusions

Approximately 2049 sets of thermal comfort data were collected throughout 2015 from Malaysia, Indonesia, Singapore, and Japan during hot and humid weather. In the investigated buildings, different ventilation modes were used, including mechanical cooling (CL), free running (FR), or mixed mode (MM). The conclusions from the field study in the four countries are as follows:

1. Most of the respondents were thermally comfortable in all the investigated locations. In Japan, more than 97% of the votes were within the TSV comfort range. In Malaysia and Indonesia, the proportions were more than 75%, except for the FR mode in

Table 16
Average clothing insulation values.

Country	Mode	Variable	I_{cl} (clo)
Malaysia	CL	Mean	0.62
	(n = 1115)	SD	0.17
Indonesia	FR	Mean	0.58
	(n = 159)	SD	0.14
	MM	Mean	0.59
	(n = 150)	SD	0.15
	CL	Mean	0.52
	(n = 91)	SD	0.13
Singapore	CL	Mean	0.57
	(n = 14)	SD	0.12
Japan	FR	Mean	0.51
	(n = 37)	S.D.	0.11
	CL	Mean	0.53
	(n = 418)	S.D.	0.14

Note: CL: Cooling; FR: Free running; MM: Mixed mode; I_{cl} : Clothing insulation; n: Number of samples; S.D.: Standard deviation.

- Indonesia, with more votes for a warm sensation. In contrast, most votes in the CL mode in Singapore were for a cool sensation.
2. Based on a probit analysis, 80% of the respondents were thermally comfortable, with the T_{op} range between 24.5 °C and 30 °C in the CL mode in Malaysia, and between 26 °C and 28 °C in MM ventilation in Indonesia. These results were actually higher than the recommended temperatures of each country.
3. Based on the Griffiths' method, the highest comfort temperatures were in Indonesia with the MM ventilation at 27.5 °C T_{op} , followed by the CL mode in Singapore, Indonesia, Japan, and Malaysia, which were 26.4 °C, 26.3 °C, 25.8 °C, and 25.6 °C, respectively. The lowest comfort temperatures were in the FR mode at 25.8 °C in Japan and 24.7 °C in Indonesia. Based on the results from Japan, the differences in the comfort temperature

could be attributed to the occupants adapting to the conditions, rather than only the type of ventilation.

4. Compared with the CEN standards, the results for the FR mode were mostly within the comfort zone. The results in MM ventilation in Indonesia were compatible with a previous study in Pakistan. The results for the CL mode in all the locations were mostly within the CIBSE comfort zone.
5. The highest mean comfort temperature in Japan occurred when the occupants felt less humid (dry condition). Relevant to air movement, the mean comfort temperatures in all the locations increased when the occupants indicated a sensation of higher air movement.
6. The most frequent adaptive behaviour in Malaysia was using AC for cooling purposes, whereas the occupants in Japan and Indonesia preferred to consume cold food or beverages. The high comfort temperature relevant to MM ventilation in Indonesia could have been affected by more options for adaptive

action, as the second most frequent action in Indonesia was opening windows and doors.

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Appendix A1

Thermal comfort questionnaire survey

Q0. Socio-demographic

- i. Gender: ☐ male ☐ female
- ii. Age: _____ years
- iii. Weight: _____ kg
- iv. Height: _____ cm
- v. Nationality: _____

Q1 – Q9. Thermal sensation, acceptability, preference and comfort

- 1) How do you feel your current health condition?
 - 4 ☐ Good
 - 3 ☐ Fair
 - 2 ☐ Not good
 - 1 ☐ Bad
- 2) How would you rate your overall comfort?
 - 6 ☐ very comfortable
 - 5 ☐ comfortable
 - 4 ☐ slightly comfortable
 - 3 ☐ slightly uncomfortable
 - 2 ☐ uncomfortable
 - 1 ☐ very uncomfortable
- 3) How do you feel about the thermal environment in your workplace right now?
 - +3 ☐ hot
 - +2 ☐ warm
 - +1 ☐ slightly warm
 - 0 ☐ neutral
 - 1 ☐ slightly cool
 - 2 ☐ cool
 - 3 ☐ cold
- 4) Is this thermal condition acceptable to you?
 - +1 ☐ acceptable
 - 1 ☐ unacceptable
- 5) What type of thermal environment would you prefer right now?
 - 2 ☐ compare to current state, I prefer much warmer
 - 1 ☐ compare to current state, I prefer slightly warmer
 - 0 ☐ prefer not to change the thermal environment
 - 1 ☐ compare to current state, I prefer slightly cooler
 - 2 ☐ compare to current state, I prefer much cooler
- 6) How do you feel about the air movement in your workplace right now?
 - 1 ☐ I do not feel the air flow at all
 - 2 ☐ I feel the flow of weak air
 - 3 ☐ I feel the flow of moderate air
 - 4 ☐ I feel the flow of strong air

- 7) Are you comfortable with the air movement in your workplace right now?
 - +1 ☐ Yes
 - 1 ☐ No

- 8) How do you feel the humidity in this room?
(If you are not sure, please compare with your skin condition. You could tell humidity is higher as your skin gets stickier)
 - +3 ☐ humidity is very high
 - +2 ☐ humidity is high
 - +1 ☐ humidity is slightly higher
 - 0 ☐ neither humid nor dry
 - 1 ☐ slightly dry
 - 2 ☐ dry
 - 3 ☐ very dry

- 9) How would you prefer the humidity to be?
 - 2 ☐ I prefer much drier than current state
 - 1 ☐ I prefer slightly drier than current state
 - 0 ☐ I do not want to change the current state
 - +1 ☐ I prefer a slightly higher humidity rather than the current state
 - +2 ☐ I prefer a humid state rather than the current state

Q10. Adaptive actions

- 10) Please tick ☒ if you **have done** the following actions as a result of feeling hot/cold **now**.
(Multiple answers are acceptable)

- a ☐ I changed the setting temperature of the air conditioner
- b ☐ I changed the speed of the fan
- c ☐ I closed or opened the doors /windows for ventilating purpose
- d ☐ I closed or opened the curtains or blinds for shading purposes
- e ☐ I changed the clothes I am wearing (took off my coat, wearing a jacket, wearing socks or shawls, etc.)
- f ☐ I drank hot or cold water or juices
- g ☐ I took a rest or doing things less vigorously
- h ☐ I changed my seating position or moved around
- i ☐ I rinsed face and hands
- j ☐ I did nothing
- k ☐ Others ()

Q11. Clothing insulation

Please tick (✓) on the following list provided, for the clothing items that you are wearing now. *In addition, for those who do not want to tick on the list, please sum up the amount of clothing as listed beside the garment list. Please fill in the clo value here. [clo].*

Classification	Clo Value	Garment's description	Deskripsi pakaian	Please tick here (✓)
Male				
Inner (<i>Pakaian Dalam</i>)	0.04	Men's brief	"Men's brief"	
	0.06	Shirt sleeveless	T-shirt tiada lengan	
	0.09	t-shirt	T-shirt nipis	
	0.10	Thin pants	Seluar nipis	
Sock (<i>Stokin</i>)	0.02	Ankle length athletic socks (thin)	Stokin paras buku lali	
	0.03	Calf-length socks	Sokin paras betis	
	0.05	Thick ankle socks	Stokin tebal paras buku lali	
	0.06	Knee socks (thick)	Stokin tebal paras lutut	
Shoes (<i>Kasut</i>)	0.02	Sandals or shoes	Sandals atau kasut	
	0.10	Boots	Boot	
Short sleeves shirt (<i>Baju Lengan pendek</i>)	0.17	Short sleeves sport shirt	Tshirt sukan lengan pendek	
	0.19	Short sleeves t-shirt	Tshirt lengan pendek	
	0.15	Short sleeves shirt	Kemeja lengan pendek	
Long sleeves shirt (<i>Baju lengan panjang</i>)	0.25	Long sleeves t-shirt	Tshirt lengan panjang	
	0.15	Long sleeves shirt	Kemeja lengan panjang	
	0.34	Long sleeves with turtleneck blouse	Tshirt lengan panjang dengan leher kolar tinggi	
	0.34	Long sleeves sweatshirt	tshirt lengan panjang tebal	
Bottom (<i>Bahagian Bawah</i>)	0.06	Shorts	Seluar pendek	
	0.11	Walking shorts	Seluar pendek ketika berjalan	
	0.15	Light trousers	Seluar slack nipis	
	0.24	Thick long trousers	Seluar jeans	
Outer (<i>Bahagian luar</i>)	0.12	Sleeveless vest	Vest tidak berlengan	
	0.20	Thin sweater	Sweater nipis	
	0.26	Jacket	Jaket	
Female				
Inner (<i>Pakaian Dalam</i>)	0.01	Pantyhose	Sarung kaki	
	0.04	Girdle	Bengkung	
	0.05	Bras & panties	Bras & panties	
	0.06	Shirt sleeveless	Tshirt nipis tiada lengan	
Sock (<i>Stokin</i>)	0.02	Ankle length socks (thin)	Stokin paras buku lali	
	0.03	Calf-length socks	Sokin paras betis	
	0.05	Thick ankle socks	Stokin tebal paras buku lali	
	0.06	Knee socks (thick)	Stokin tebal paras lutut	
Shoes (<i>Kasut</i>)	0.02	Sandals/shoes	Sandals atau kasut	
	0.10	Boots	Boot	
Short sleeves shirt (<i>Baju Lengan pendek</i>)	0.17	Short sleeves sport shirt	Tshirt sukan lengan pendek	
	0.19	Short sleeves t-shirt	Tshirt lengan pendek	
Long sleeves shirt (<i>Baju lengan panjang</i>)	0.24	Work jacket, female	Jaket bekerja wanita	
	0.25	Long sleeves t-shirt	Tshirt lengan panjang	
	0.34	Long sleeves with turtleneck blouse	Tshirt lengan panjang dengan leher kolar tinggi	
	0.34	Long sleeves sweatshirt	Tshirt lengan panjang tebal	
	0.15	Long sleeves shirt	Kemeja lengan panjang	
Bottom (<i>Bahagian Bawah</i>)	0.06	Shorts	Seluar pendek	
	0.11	Walking shorts	Seluar pendek ketika berjalan	
	0.15	Thin long trousers	Seluar slack nipis	
	0.24	Thick long trousers	Seluar jeans	
Bottom skirt (<i>Skirt</i>)	0.5	Baju Kurung	Baju Kurung	
	0.14	Thick skirt knee length	Skirt tebal paras lutut	
	0.23	Thick skirt ankle length	Skirt tebal paras buku lali	
	0.36	One piece	"one piece"	
Outer (<i>Bahagian luar</i>)	0.25	Long sleeves sweater (thin)	Sweater lengan panjang (nipis)	
	0.26	Jacket	Jaket	
	0.05	Headress/Hijab	Tudung/Hijab	

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